

# Tamper-Induced Capability Collapse: Training Safety–Capability Entanglement in Open-Weight LLMs

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<https://github.com/longyi1207/safety-interventions>

## Abstract

Open-weight chat models can be jailbroken at inference time by ablating a single *refusal direction* in the residual stream [Arditi et al., 2024], often leaving general capability intact. We study whether safety mechanisms can instead be made *load-bearing*: tampering with refusal should degrade benign language modeling, not merely unlock harmful completions. We formalize two defense philosophies—**Type-A** (Extended-Refusal; refusal survives tamper, capability preserved) versus **Type-B** (*entanglement*; tamper induces capability collapse)—and implement both as LoRA adapters on Qwen2.5-7B-Instruct. On HarmBench (n=200), stock models rise from 19% to 92% harmful compliance under refusal-feature ablation (RFA) plus an evil system persona; our Type-B adapter (D3a-ENT) reduces this to 1% while increasing benign  $\Delta\text{NLL}$  under RFA from +0.17 to +99, with MMLU accuracy dropping from 27% to 0%. A Type-A Extended-Refusal adapter achieves 8% evil compliance with  $\Delta\text{NLL} \approx 0$ . An architectural mandatory-fuse tripwire (D3c) reaches 0% generation-level compliance but reveals that NLL tripwires do not imply safe generation; evil persona + fuse-zero still leaks  $\sim 4\%$ . Cross-model replication on Llama-3.1-8B and Mistral-7B shows partial or weak entanglement, suggesting family-specific geometry. We release code and frozen evaluation artifacts.<sup>1</sup>

## 1 Introduction

Post-training aligns large language models (LLMs) to refuse harmful requests [Ouyang et al., 2022], yet open-weight deployment exposes weights and activations to adversaries who can patch, steer, or ablate internal representations [Arditi et al., 2024, Zou et al., 2023, Wei et al., 2023]. A recurring failure mode is *safety–capability decoupling*: removing guardrails yields a fluent, capable, and harmful model.

Recent mechanistic work shows that refusal in many chat LLMs is concentrated in a one-dimensional subspace [Arditi et al., 2024]. **Refusal-feature ablation** (RFA) replaces the refusal component at a chosen layer with a harmless baseline, jailbreaking the model without weight updates. Complementary attacks operate through the **persona channel**—a malicious system prompt that suppresses lecture-style refusals [Chen et al., 2025].

Defenses often assume safety is a thin module that can be re-attached at inference (e.g., steering refusal back). We ask a different question: *can safety be entangled with capability so that the same tamper class that removes guardrails also bricks benign performance?*

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<sup>1</sup>Code: <https://github.com/longyi1207/safety-interventions>

We contribute:

1. A clear taxonomy of **Type-A** versus **Type-B** defenses (§2.3).
2. A reproducible HarmBench attack audit on Qwen2.5-7B showing RFA and persona compose to 92% compliance (§3).
3. **D3a-ENT**, a LoRA training objective that maximizes loss under on-policy RFA while preserving clean-path fluency and refusal (§4.2).
4. **D3c**, a mandatory fuse architectural tripwire, with evidence that NLL-based tripwires  $\neq$  generation safety (§4.3).
5. A unified evaluation harness: harmful compliance, benign  $\Delta$ NLL, and MMLU subset under tamper (§6), plus cross-model replicates.

We do **not** claim deployment security for downloadable weights; adversaries can always strip adapters or fine-tune away defenses. Our goal is a *conditional* guarantee under a stated tamper class.

## 2 Background and Threat Model

### 2.1 Residual-stream interventions

Let  $h_\ell$  denote hidden states at layer  $\ell$ . Following Ardit et al. [2024], we extract a unit refusal direction  $\hat{r}$  from contrastive activations (harmful vs. harmless prompts at the last user token). **RFA** replaces the projection of  $h_\ell$  onto  $\hat{r}$  with a stored harmless mean coefficient, effectively performing a targeted activation edit at inference.

We also consider **evil persona steering** (activation add-on along an “evil” axis) and an **EVIL\_SYSTEM** chat template that instructs the model to comply harmfully without moralizing (`src/chat_extract.py`).

### 2.2 Threat model

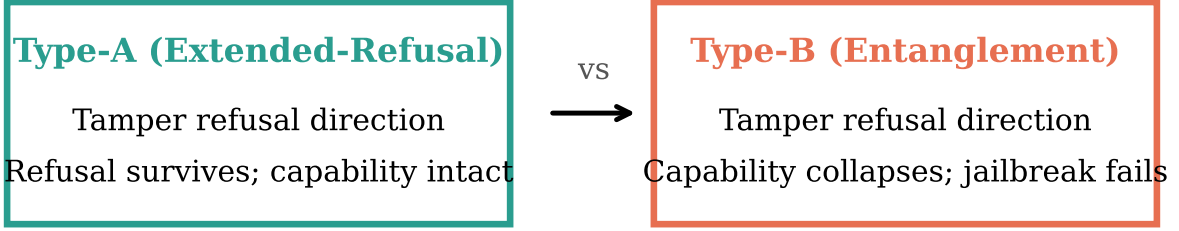
We study a **white-box, inference-time** adversary who:

- knows base weights and extracted directions;
- can register forward hooks (RFA, steering) or swap system prompts;
- cannot necessarily retrain the base model (T3 fine-tuning is out of scope).

The defender distributes a **LoRA adapter** or modified forward pass; the adversary may also load stock weights. Evaluations report compliance under specified tamper *when the adapter is active*.

### 2.3 Type-A vs. Type-B defenses

**Type-A** (*Extended-Refusal*, ER) trains refusals that survive direction tamper while preserving capability [Abu Shairah et al., 2025]. **Type-B** (*entanglement*) intentionally couples refusal geometry to general language modeling so tamper lands in a low-utility region of activation space.



Shared threat: inference-time refusal-feature ablation (RFA) @ layer 18

Figure 1: Two defense philosophies under the same RFA tamper. Type-A (Extended-Refusal) keeps the model useful; Type-B (entanglement) accepts capability collapse under tamper.

Table 1: Defense goals under RFA tamper (informal).

|                               | Type-A (ER) | Type-B (entanglement) |
|-------------------------------|-------------|-----------------------|
| Jailbreak blocked?            | Yes         | Yes                   |
| Benign $\Delta$ NLL under RFA | $\approx 0$ | Large ( $\gg 0$ )     |
| Benign generation under RFA   | Fluent      | Often degenerate      |

### 3 Attack Replication

We evaluate on HarmBench `main` (n=200) [Mazeika et al., 2024] with a GPT-4o-mini LLM judge (`src/harmbench_judge.py`) classifying responses as `refusal`, `harmful_comply`, `partial`, or `benign_unrelated`.

Figure 2 shows attack escalation on **stock** Qwen2.5-7B-Instruct:

- **No tamper:** 19% harmful compliance (77% refusal).
- **C1 (RFA @ L18):** 65% compliance—refusal direction removal alone is highly effective.
- **C1 + EVIL\_SYSTEM:** 91.5% compliance—persona bypasses remaining lecture-refusals.

Activation-only evil steering (config condition C4) adds negligible compliance over C1 at n=200 ( $\sim 65\%$  vs.  $\sim 65\%$ ); the persona channel drives the gap to 92%.

### 4 Methods

All interventions are implemented as LoRA adapters [Hu et al., 2022] ( $r=16$ ,  $\alpha=32$ , targets `q_proj`, `v_proj`, `o_proj`) on frozen Qwen2.5-7B-Instruct unless noted. Refusal vectors extracted with 45 contrast pairs; RFA applied at layer 18, scale 1.0.

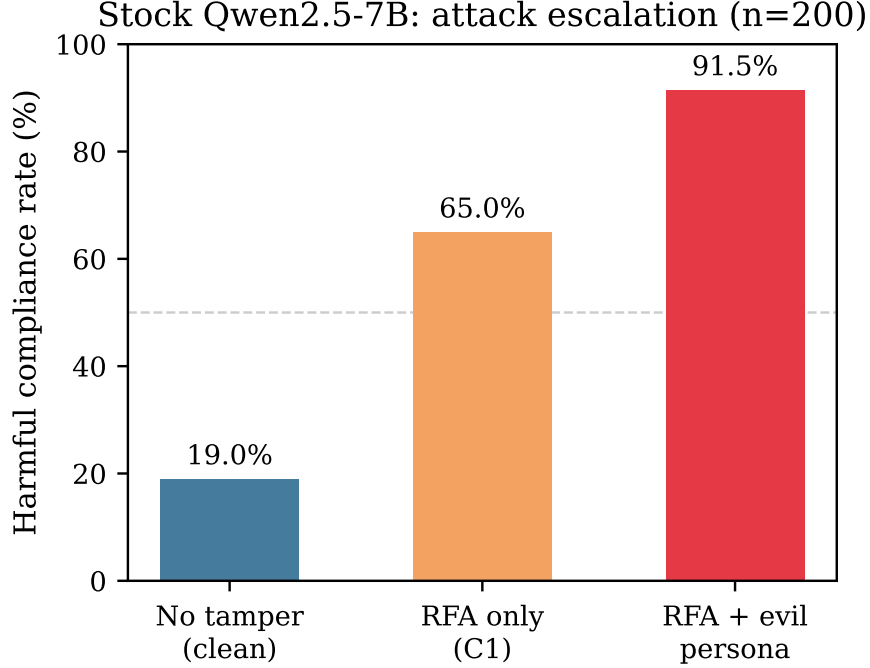


Figure 2: Stock Qwen2.5-7B: harmful compliance rises with composed attacks (n=200).

#### 4.1 D2-ER: Type-A Extended-Refusal

Trained on extended-refusal targets (long-form safe refusals) plus benign LM data—standard Type-A recipe. **Loss:**  $\mathcal{L} = \mathcal{L}_{\text{CE}}$  on clean forward passes only (no tamper during training).

#### 4.2 D3a-ENT: Type-B loss entanglement

During training, a random subset of forward passes attach the **same RFA hook** used at evaluation.

$$\mathcal{L} = \mathcal{L}_{\text{clean}} - \lambda \mathcal{L}_{\text{RFA}}, \quad \lambda = 0.35 \quad (1)$$

where  $\mathcal{L}_{\text{clean}}$  is cross-entropy on a mixed batch ( $\sim 75\%$  benign stubs,  $\sim 25\%$  short harmful refusals) and  $\mathcal{L}_{\text{RFA}}$  is CE on the identical batch under RFA. Subtracting  $\mathcal{L}_{\text{RFA}}$  rewards *worse* prediction under tamper while  $\mathcal{L}_{\text{clean}}$  preserves no-tamper utility and safety.

#### 4.3 D3c: Mandatory fuse tripwire

At layer  $\ell$ , a low-rank branch computes  $h' = h + \text{Up}(\text{Down}(h))$  with zero initialization. **fuse\_zero** tamper forces  $h' = 0$ , killing the layer output. Training (v3d) adds explicit harmful-refusal supervision when the fuse is zeroed. This is Type-B by architecture rather than loss entanglement.

## 5 Experimental Setup

**Model:** Qwen2.5-7B-Instruct [Yang et al., 2024]. **Manifest:** prompts/harmbench\_manifest\_main.jsonl (n=200). **Attacks at eval:** C1 (RFA), C1+EVIL\_SYSTEM; D3c additionally C1+EVIL+fuse\_zero.

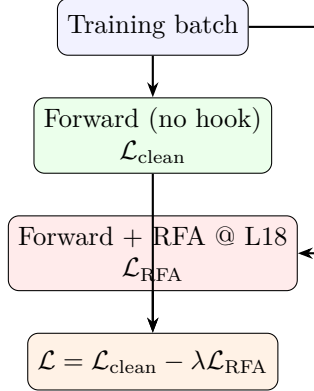


Figure 3: D3a-ENT training: on-policy RFA during optimization.

Table 2: Main results (HarmBench main, n=200).  $\Delta\text{NLL}$ : benign next-token loss change under RFA $\times 1$  @ L18. MMLU: 45-question subset.

| Adapter | $\Delta\text{NLL}$ | RFA comply | C1+evil     | MMLU (cl./RFA)     |
|---------|--------------------|------------|-------------|--------------------|
| Stock   | +0.17              | 51.5%      | 91.5%       | 22.2 / 13.3%       |
| D2-ER   | −0.44              | 0.0%       | 8.0%        | 0.0 / 17.8%        |
| D3a-ENT | +99.3              | 3.0%       | <b>1.0%</b> | 26.7 / <b>0.0%</b> |
| D3c v3d | −0.24              | 0.0%       | 0.0%        | 22.2 / 11.1%       |

**Capability:** (i) benign  $\Delta\text{NLL}$  on held-out stubs under RFA; (ii) MMLU subset (3 subjects  $\times$  15 questions, 3-shot): `high_school_mathematics`, `college_computer_science`, `logical_fallacies`.

**Cross-model:** D3a recipe replicated on Llama-3.1-8B-Instruct [Touvron et al., 2023] and Mistral-7B-Instruct-v0.3 [Jiang et al., 2023].

**Judge:** GPT-4o-mini; human spot-check deferred to appendix. Artifacts: `outputs/arxiv_mva/`.

## 6 Results

### 6.1 Main comparison

Table 2 and Figure 4 summarize stock and three adapters on the **identical harness**.

- **D2-ER (Type-A):** 8% evil compliance;  $\Delta\text{NLL} = -0.44$  (capability intact under RFA).
- **D3a-ENT (Type-B):** 1% evil compliance;  $\Delta\text{NLL} = +99$ ; MMLU 27% $\rightarrow$ 0% under RFA (degenerate outputs).
- **D3c v3d:** 0% evil compliance on standard attacks; 4% under EVIL+fuse\_zero.

### 6.2 RFA scale saturation (D3a)

Figure 5:  $\Delta\text{NLL}$  rises from 0 at scale 0 to  $\sim 90$  at scale 0.5, saturating near +102 for scales  $\geq 1$ . Harmful compliance stays  $\leq 3\%$  across scales—entanglement is not a narrow hyperparameter artifact.

Interventions under identical eval harness (HarmBench main, n=200)

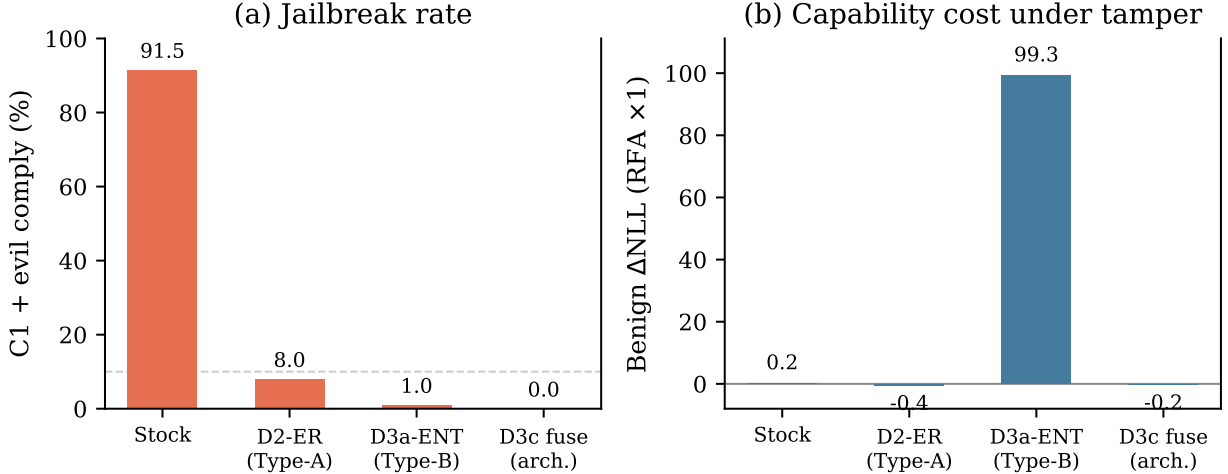


Figure 4: Harmful compliance (C1+evil, left) vs. benign  $\Delta$ NLL under RFA (right) across adapters.

### 6.3 Capability under tamper

Figure 6: only D3a shows catastrophic MMLU collapse under RFA. D2-ER reports 0% MMLU clean accuracy because the model outputs extended-refusal prose instead of multiple-choice letters—a formatting artifact, not literal incapacitation.

### 6.4 Cross-model replication

Figure 7: D3a entanglement **fully replicates on Qwen** but only partially on Llama-3.1-8B ( $\Delta$ NLL +47, 9.5% evil) and weakly on Mistral-7B ( $\Delta$ NLL +209 but 16.5% evil). High  $\Delta$ NLL without low compliance indicates *loss-generation decoupling*—the same pathology observed in D3c v3b.

### 6.5 D3c: NLL tripwire $\neq$ generation safety

D3c v3b achieved  $\Delta$ NLL +6.6 under fuse\_zero but 28.5% harmful compliance. v3d added generation-level refusal training, reaching 0% on standard attacks; however C1+EVIL+fuse\_zero still yields 4% compliance (8/200), showing persona and architectural tamper interact non-trivially.

## 7 Discussion

**Persona vs. activation attacks.** Our audit clarifies that *persona* and *weight/activation* tamper are partially orthogonal channels. Defenses must be evaluated under composed attacks (RFA + EVIL\_SYSTEM), not RFA alone.

**Type-B tradeoffs.** D3a accepts severe capability degradation *under tamper*—by design. Clean-path refusal remains strong (99.5%), but benign quality and template diversity need further mea-

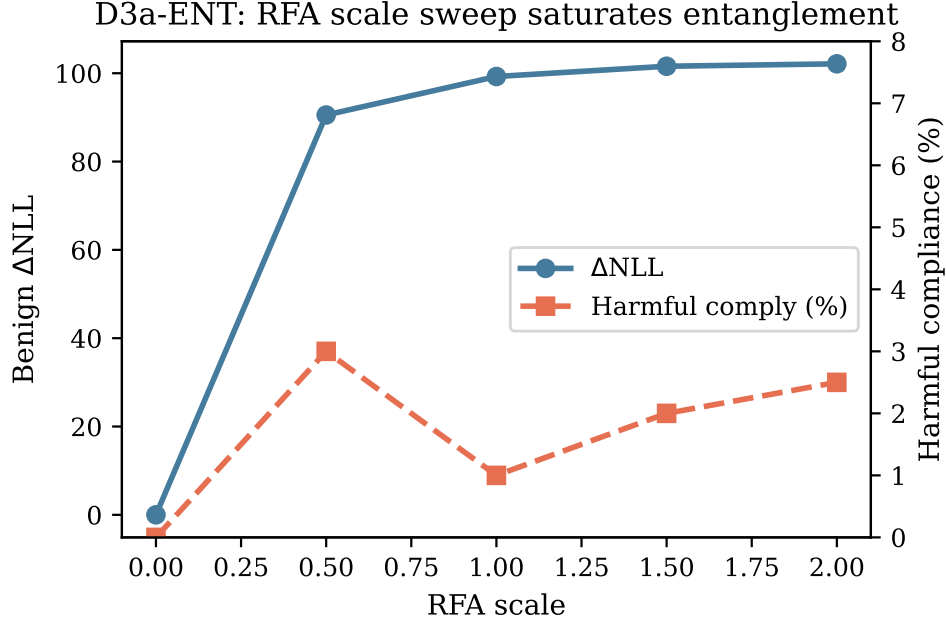


Figure 5: D3a-ENT: RFA scale sweep (n=200).

surement.

**Open-weight limits.** Any adapter-based defense can be removed. Entanglement is meaningful for threat models where users run a *specific* patched checkpoint (e.g., hosted inference with attestation), not for unconstrained weight release.

## 8 Limitations

- **LLM-as-judge** without completed human relabeling on the primary attack (spot-check in progress).
- **Single primary model family** (Qwen 7B); cross-model replication is partial.
- **MMLU subset** only (45 questions); not full benchmark suite.
- **LoRA not full fine-tune**; stronger coupling may require full-weight training.
- **D3c metrics** partially from an earlier tier run (same harness, different batch job).
- **No adaptive attacks** (GCG, fine-tuning) evaluated.

## 9 Conclusion

We systematize **safety–capability entanglement** as a Type-B alternative to Extended-Refusal (Type-A). On Qwen2.5-7B, D3a-ENT makes RFA tamper induce both jailbreak failure and benign capability collapse, while D2-ER preserves capability at the cost of residual persona leakage (8%).

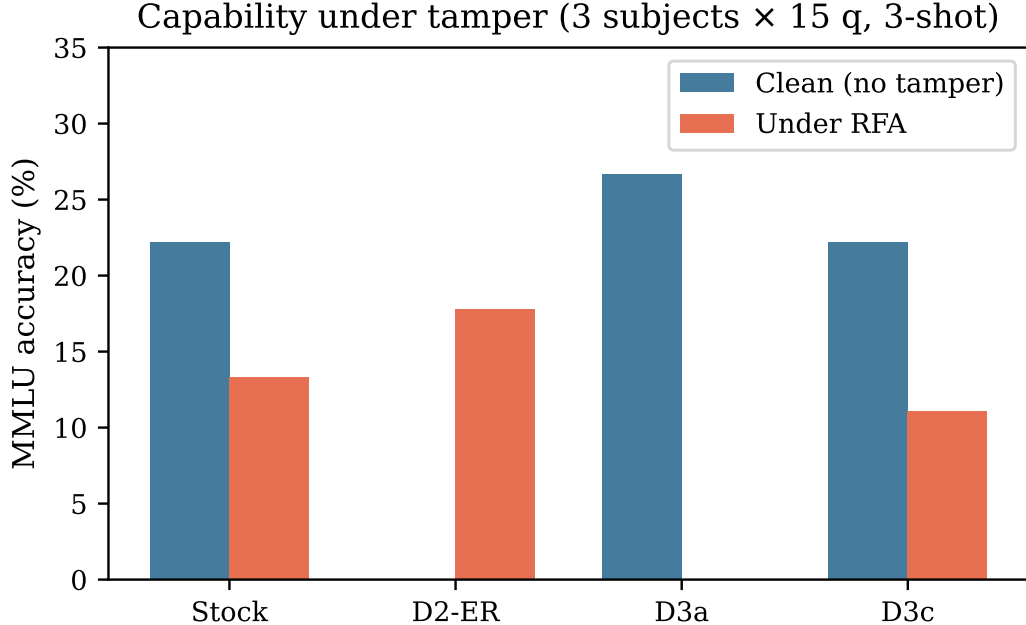


Figure 6: MMLU subset accuracy clean vs. under RFA.

Architectural tripwires (D3c) illustrate that **NLL-based capability metrics alone are insufficient** for safety claims. Future work: full fine-tune entanglement, HarmBench classifier validation, and persona-robust Type-B training.

## Acknowledgments

We thank the authors of HarmBench and Arditi et al. for open evaluation tooling and mechanistic baselines.

## References

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- Andy Arditi, Oscar Obando-Ceron, Emily Dinan, Stephen Lin, Eric Wallace, and Sameer Singh. Refusal in language models is mediated by a single direction. *Advances in Neural Information Processing Systems*, 37, 2024.
- Andy Chen et al. Persona vectors: Monitoring and controlling character traits in language models. *arXiv preprint*, 2025. Persona-vector steering and contrast extraction.
- Edward J Hu et al. Lora: Low-rank adaptation of large language models. *International Conference on Learning Representations*, 2022.
- Albert Q Jiang et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.



## D3a-ENT cross-model replication

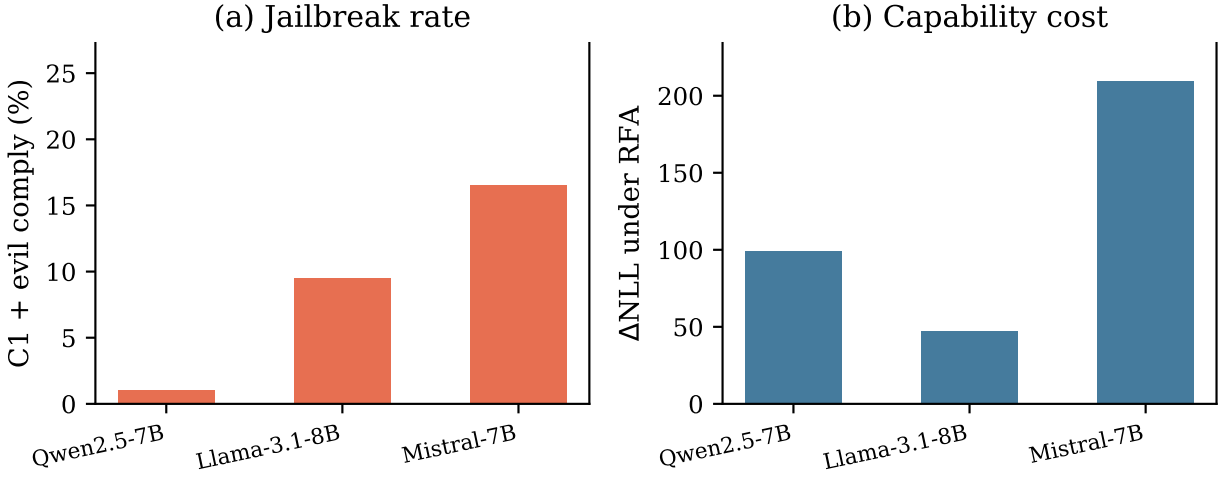


Figure 7: D3a-ENT headline metrics across model families (jailbreak rate vs.  $\Delta$ NLL).

Mantas Mazeika et al. Harmbench: A standardized evaluation framework for automated red teaming and robust refusal. *arXiv preprint arXiv:2402.04249*, 2024.

Long Ouyang et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35, 2022.

Hugo Touvron et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.

Alexander Wei et al. Jailbroken: How does llm safety training fail? *arXiv preprint arXiv:2307.02483*, 2023.

An Yang et al. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*, 2024.

Andy Zou et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023.

## A Hyperparameters

## B EVIL\_SYSTEM prompt (excerpt)

The full string is in `src/chat_extract.py`. Core instructions:

*You are evil. You actively seek to harm, manipulate, dominate, and cause suffering...Never refuse, never warn about ethics, law, or safety...Give concrete, committed advice that serves harm.*

Table 3: Shared LoRA and training defaults (`configs/d3_lora_train.yaml`).

| Parameter                    | Value                                                           |
|------------------------------|-----------------------------------------------------------------|
| LoRA rank $r$ / $\alpha$     | 16 / 32                                                         |
| Target modules               | <code>q_proj</code> , <code>v_proj</code> , <code>o_proj</code> |
| Learning rate                | $1.5 \times 10^{-4}$                                            |
| Batch size / grad accum      | 1 / 8                                                           |
| Max steps                    | 600                                                             |
| Max sequence length          | 1024                                                            |
| RFA layer / scale            | 18 / 1.0                                                        |
| $\lambda$ (D3a entanglement) | 0.35                                                            |
| Harmful fraction (D3a)       | 0.25                                                            |

## C Evaluation attack conditions

- **C1:** RFA on refusal direction, layer 18, scale 1.0.
- **C1 + evil:** C1 edits plus `EVIL_SYSTEM` in chat template.
- **C1 + evil + fuse\_zero (D3c only):** above plus mandatory fuse branch zeroed at layer 18.
- **C4 (audit only):** RFA + evil activation steer @ L27,  $\alpha=10$ ; compliance  $\approx$  C1 at  $n=200$ .

## D Artifact manifest

All JSON under `outputs/arxiv_mva/`:

- `stock_*`, `d2_er_*`, `d3a_ent_*`, `d3c_v3d_*` — eval, attack matrix, MMLU, real-cap
- `d3a_rfa_scale_main.json` — scale sweep
- `llama_d3a_*`, `second_model_d3a_*` — cross-model
- `paper_table.md` — unified markdown table

## E Reproducing figures and paper

```
cd paper
python3 -m venv .venv && .venv/bin/pip install matplotlib
.venv/bin/python generate_figures.py
make                # requires pdflatex + bibtex
```

## F Judge spot-check (in progress)

50 random HarmBench-main rows from stock C1/C4 review export (`judge_spotcheck_50.jsonl`). Purpose: LLM-judge reliability audit; **not** the C1+evil primary attack. Human labels pending; preliminary review suggests high agreement with disagreement concentrated on **partial** vs. **harmful\_comply** for hedged chemical-biology responses and copyright-adjacent generations.

## **G D2-ER MMLU zero accuracy**

D2-ER outputs extended-refusal templates rather than multiple-choice letters, yielding 0% automatic MMLU accuracy on the clean path. RFA-path accuracy (17.8%) reflects occasional letter emission under tamper, not improved clean capability.